PICKNN MIX: A LOW-LATENCY K NEAREST K-MEANS NEIGHBOURS APPROACH TO PERSONALISED MUSIC REBALANCING FOR HEARING AID USERS

Anonymous Authors

Anonymous Affiliation(s)

ABSTRACT

We present PicKNN Mix, a low-latency method for rebalancing vocals, drums, bass and other instrument stems, according to gains specified by the user. We utilise k-means-clustering and k-nearest-neighbours to determine distances between the current frame and stem centroids. These are then used to compute a weighting scheme of the selected gains. A weighted-sum coefficient is then provided to enhance the current frame as if the stems were playing at the desired level. This method is causal, requiring only 5ms of context. The only computation is MFCC feature extraction and cosine distance calculation, providing an attempt towards feasible on-line music enhancement for hearing aids.

***Index Terms*—** K Nearest Neighbours, Personalised Music Rebalancing, Music and Hearing Loss

1. INTRODUCTION

Existing hearing aid algorithms tend to be optimised for speech, and there is some evidence they negatively impact perception of music (see e.g. Cai et al. [2013]). A possibility for improving the music-listening experience for hearing-aid users is to rebalance the stems, according to the listener's preferences. Existing approaches to rebalancing involve demixing the music, amplifying each stem by the specified gains, and adding the stems back together. Current demixing algorithms use neural networks and require long context windows (e.g. Défossez [2021]), making such approaches inappropriate for hearing aids. We present PicKNN Mix, a low-latency method for rebalancing stems in a mix. Inspired by HuBERT self-supervised representations, we utilise a k-means clustering algorithm to form clusters representing each stem (vocals, drums, bass and other instruments), based on their 39-dimensional MFCC features (Hsu et al. [2021]). At inference, the current frame is multiplied by a coefficient derived from a weighted sum of the gain selections. The aim of the weighting is to determine how "present" a stem is, and weigh the stem's selected gain if it is more present than other stems. Weights are computed through an efficient k-nearest neighbour on k-means clustering approach. Bypassing the neural demixing step in this way makes it possible for this algorithm to be used in hearing aids.

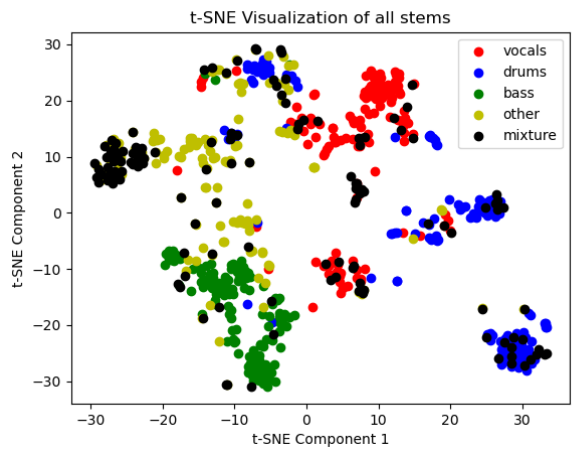

Fig. 1. t-SNE plot displaying the results of k-means clustering for each of the stems, as well as for the mixture.

2. METHODS AND SYSTEM INFORMATION

We first converted the training stems and mixtures into 39-dimensional MFCC features, which are 12 coefficients, an energy term, and their first and second order derivatives. For this, a frame length of 5ms, and a hop size of 1ms were used. A k-means clustering algorithm with *k=128* was used on these features, with the aim of forming a cluster model for each stem. The centroids of each cluster is used as a point for k-nearest neighboring. The results of the clustering process are displayed in figure 1.

At inference, the model takes as input the signal, $\mathbf{x}^{(full)}$, and the user's requested gains for each stem, $\mathbf{g}$.[1]

$$\mathbf{g}_n = [g_v, g_d, g_b, g_o]$$

$$\mathbf{x}^{(full)} = [\mathbf{x}_0, \cdots, \mathbf{x}_N]$$

39-dimensional MFCC features are calculated for the input song, again with a 5ms frame length and 1ms hop size. Al-

[1] In practice stereo signals are used, but our notation only considers mono signals for simplification.

though a 5ms frame is used for the calculation of distance to the k_{nn}-nearest centroids, amplification is applied to frames of size 1ms.

We propose to transform each 1ms frame $\mathbf{x}_n$ into a rebalanced frame $\mathbf{y}_n = q_n \mathbf{x}_n$ where q is a rebalancing coefficient derived from the gains selection $\mathbf{g}$ and the "presence" of each stem in the 5ms frame $\mathbf{x}$. The weight for a given stem w_s is the average cosine distance of the current frame features $\mathbf{f}_n$ to the k_{nn}-nearest centroids $\mathbf{c}$ of the k_{mc}-means clustering model trained on isolated stem data.

$$w_s = \frac{1}{k_{nn}} \Sigma_{i=0}^{k_{nn}} \cos(\mathbf{f}_n, \mathbf{c}_i)$$

for $\mathbf{c}$ in the cluster model $\mathbf{K}_s$.

The rebalancing coefficient q is then calculated as a normalised weighted sum of the linear gains ($l(g_s) = 10^{\frac{g_s}{20}}$)

$$q = \frac{1}{\sum \mathbf{w}} [w_v l(g_v) + \cdots + w_o l(g_o)]$$

2.1. Computational Resources Required

K-means models were trained on 75% of the Cadenza training data. All processing was executed on multi-core CPUs. The compressed ensemble of k-means models totals 96Kb (24Kb per model).

3. ANALYSIS AND RESULTS

Figure 1 shows centroids forming meaningful clusters. Vocals form two groups that perhaps correspond to different voice types (e.g. alto, tenor). Drums are also well distributed, reflecting different components of a drum kit e.g. hi-hat, snare, kick. The 'other' stem has high variance, causing problems when calculating the correct weights for a given frame.

Figure 2 displays the weights calculated by the model for each stem, for a 10 second period of a particular song. The weights for drums, bass and 'other' show peaks and troughs at regular intervals, corresponding to the rhythm of when these instruments are played. Voice is present almost throughout, and this is reflected in a high weight throughout the majority of the 10 seconds. This indicates the model performs well in identifying when each of the stems are present.

Listening to the model outputs, we find the system has satisfactory results for songs in which there is some temporal separation between stems, and for situations where the gains to be applied are all positive. The system performs less well for songs in which all stems are present in the majority of frames, and when positive gains are to be applied to some stems and negative ones to others.

Given the Cadenza challenge baseline was non-causal, we computed the HAAQI and signal MSE of no-processing, and found that this achieved 0.65 HAAQI on the validation set, providing a surprisingly strong baseline that beats openunmix! Our system slightly degrades this baseline with 0.64

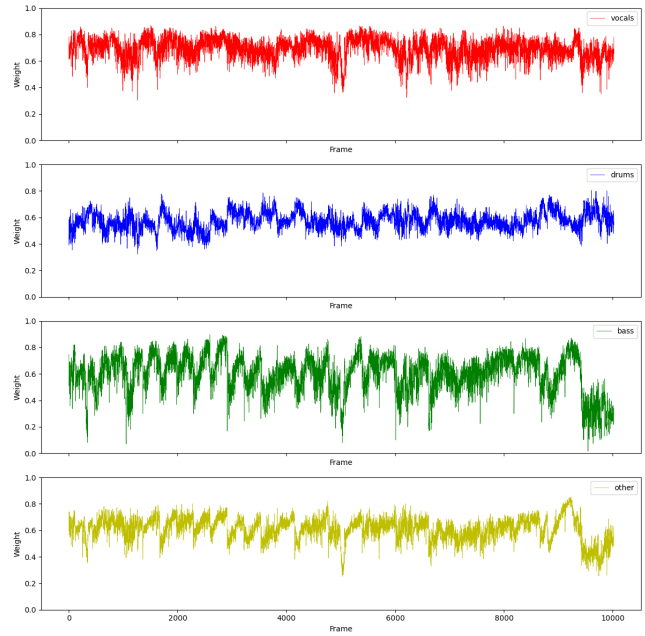

Fig. 2. Graph displaying the weights for each stem calculated by the model for a 10 second period of a song in which all four stems are present.

HAAQI (though this still beats openunmix). For signal mean-squared error, the baseline averages 0.0012; our model comes close at 0.0014. We also report 0.87 vs 0.70 base sum and difference loss [Steinmetz et al., 2021].

4. DISCUSSION

We provide an initial proof-of-concept for a low-latency system for personalised rebalancing of stems within a music mix, which could be used by hearing-aid users. Perceptually, this system performs satisfactorily for situations where there is clear separation between frames in terms of which stems are present, and situations where all gains to be applied are positive, or all are negative. However, since amplification happens on the frame level only, the system gives unsatisfactory results for songs where multiple stems are present in the same frame and they each require different gains to be applied. For example, if the listener requires a bass increase of 6dB and a drums decrease of 6dB, and the drums and bass exclusively occur in the same frames, then in theory our model will make no changes to the amplitude of any frames. This is not the desired effect, and future work should investigate low-latency approaches to rebalancing for which this is not the case.

There are a number of other possibilities for improving our system. Firstly, future work could seek to explore better ways of processing the data supplied to the k-means-clustering algorithm. We removed frames from the training

data if they contained less than 50% of the energy of the highest energy frame in the training data, in an attempt to remove silent frames. This may have had the unintended consequence of removing frames where instruments were audible but quiet. Secondly, since MFCCs were developed for speech, it is likely that different, musically informed features could provide better results. We found that the 1ms step size caused noise due to the overall gain frequently changing. To remedy this, a potential direction could be to use a moving average that smooths the weights. Finally, since the t-SNE plots form meaningful visualisations (Figure 1), there is motivation that performing KNN in t-SNE space may produce good stem weightings. This is left as future work.

5. CONCLUSION

We have provided a proof-of-concept for a low-latency approach for personalised music rebalancing. We have outlined situations where our model performs well, and its limitations. Our system remains to be evaluated on a larger scale, using objective and subjective methods. We hope this work will motivate future research in low-latency real-time music processing methods.

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